Referee Report for Paolo Pacchiotti's Essay

Grade 30/30

The paper offers a clear and well-informed comparison of Hermann's and Cassirer's responses to the problem of causality in quantum mechanics. I recommend the following improvements:

1. **Clarify the central thesis.** The introduction should state what the comparison establishes, beyond identifying similarities and differences. What does it reveal about their different ways of defending a transcendental approach?

2. **Reconsider the starting point and structure.** Begin with Hermann's and Cassirer's interpretations of Kant rather than attempting to establish Kant's own position independently, which raises controversial questions beyond the paper's scope. Follow the chronological order—Hermann first, then Cassirer—especially because Cassirer cites Hermann, whereas her contributions discussed here precede his. Use the comparative discussion as the conclusion.

3. **Qualify the characterization of Cassirer as a conventionalist.** The description of his approach as a weakening of the *a priori* "along conventionalist lines" requires textual support. In my view, this is Frank's "accusation" against Cassirer rather than Cassirer's own position: on Frank's reading, Cassirer's definition of causality ultimately amounts to a convention stipulating that the laws of nature cannot be arbitrarily complicated. Frank's charge of tautology against Hermann also deserves further discussion: what prevents retrospective explanation from merely introducing whatever cause would explain the observed effect?

4. **Develop the problem of individuality.** Although substantiality is briefly mentioned and set aside in a footnote, this issue is, in my view, central to the comparison. For Cassirer, the deeper challenge posed by quantum mechanics concerns the individuality of microscopic systems rather than the validity of causality. Hermann's rejection of this solution should be examined alongside her account of measurement.

5. **Improve the precision of some technical points.** Distinguish agreement between a mean and an individual value from the absence of dispersion. In discussing von Neumann's proof, distinguish additivity for quantum expectation values from its disputed extension to hypothetical dispersion-free subensembles.

The paper uses an author–date referencing system broadly compatible with Chicago style, but its citation formatting and typography require consistent revision:

- **Quotation marks and punctuation.** Replace guillemets with double quotation marks, using single quotation marks for quotations within quotations. In the references, periods currently appear outside closing quotation marks; Chicago places them inside.
- **Original publication dates.** In-text citations need a space between the original publication year and the year of the edition cited: (Cassirer [1937] 1956, 64–65). In the reference list, Chicago's convention uses parentheses around the original year: Cassirer, E. (1937) 1956. The paper currently uses square brackets in both places.
- **Page ranges.** Hyphens and en dashes are mixed. Use en dashes consistently: 70–80 rather than 70-80, with Chicago's preferred shortening where appropriate, such as 223–37.
- **English title capitalization.** English titles should consistently use title case. Non-English titles should follow the conventions of their respective languages.